Poisson processes

Tutorial 1

Monday 7 February

Poisson process

General Idea

" Counting process "

Count events, that happen:

— **randomly** in time (or space);

— completely **independently** from each other;

— with a **constant rate**.

Poisson process

Examples

Supermarket (your local AH) at low hours

On average, **1 customer arrives every 2 minutes**

The arrival times of the customers are:

— random
— independent (from times of previous customers)
— with constant rate of 1 customer every 2 minutes

Number of customers that have arrived during a certain time?

Poisson process

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Poisson process

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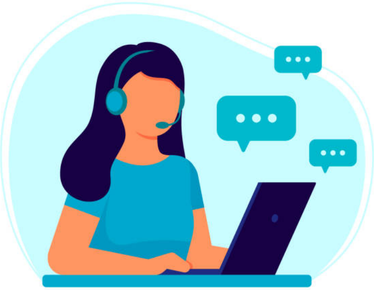
Rem: The **rate** at which customers arrive **can** also **change** in time. $\longmapsto$ " Non-homogeneous Poisson process "

Constant rate $\longmapsto$ " Homogeneous Poisson process "

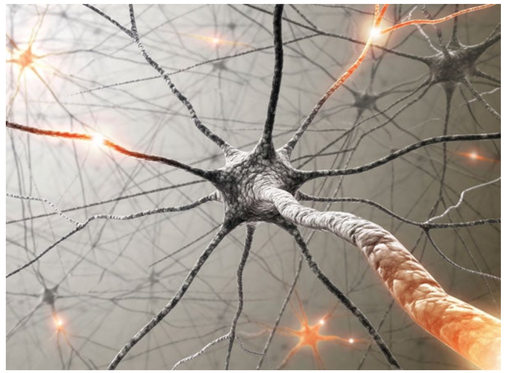
Poisson process

Used to model many other phenomena:

Arrival of customers at a supermarket



Timing of
calls at a help center



Activity of a neuron

Occurrences of earthquakes

Radioactive decay in atoms

Timing of patients
arriving in emergency room

Occurrences of meteors hitting Earth

Occurrences of
neutron-nucleus collisions
in a nuclear reactors

Location of trees in a forest

Movements in a stock price

Etc. Etc. ...

Not a Poisson process!

INDEPENDENCE between events is very important

Ex. Occurrences of earthquakes of large magnitudes

Occurrence of earthquakes of large magnitude (>5) in a country may not follow a Poisson process, if one large earthquake increases the probability of aftershocks of similar magnitude.

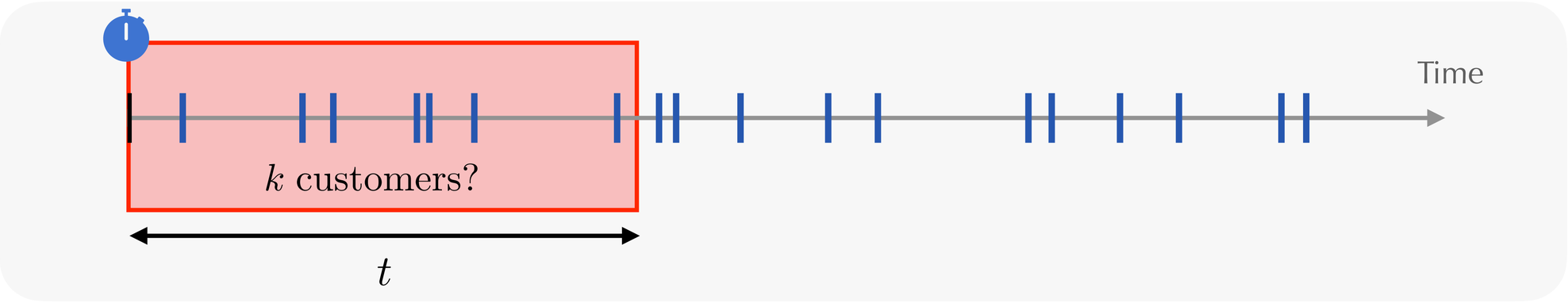
If events are random,
what can we say?

Back at our local AH

— with constant rate of 1 customer every 2 minutes

Poisson distribution

What is the probability that 10 customers have arrived in 5 min?



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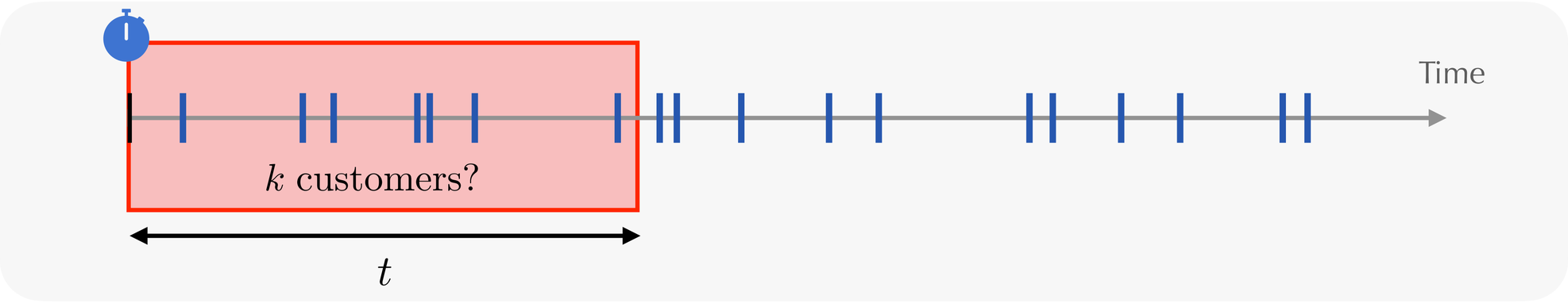
French mathematician Siméon Denis Poisson,



The counting process $N(t)$ is a **Poisson process** with rate $\lambda > 0$, if:

- $N(0) = 0$
- $N(t)$ has independent increments;
- The probability to observe k events during a time interval t is given by

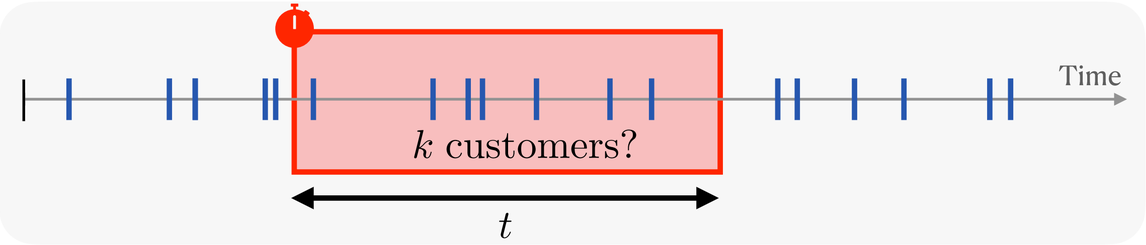
the **Poisson distribution** with parameter $\mu = \lambda t$

$$P[N(t) = k] = \frac{\mu^k \exp(-\mu)}{k!}$$

What is the probability that 10 customers have arrived in 5 min?

Rate of 1 customer every 2 minutes:

$$\lambda = \frac{1 \text{ customer}}{2 \text{ min}} = 0.5 \text{ customer per minute}$$



On average, there are 2.5 customers arriving in t=5 mins:

$$\mu = \lambda t = \frac{1 \text{ customer}}{2 \text{ min}} \times 5 \text{ min} = 2.5 \text{ customers}$$

The **probability** that **10 customers arrive in 5 mins** is given by:

$$P[N(t=5)=10] = \frac{2.5^{10} \exp(-2.5)}{10!} \simeq 0.0002$$

(Very unlikely to happen)

Poisson distribution with parameter $\mu = \lambda t$

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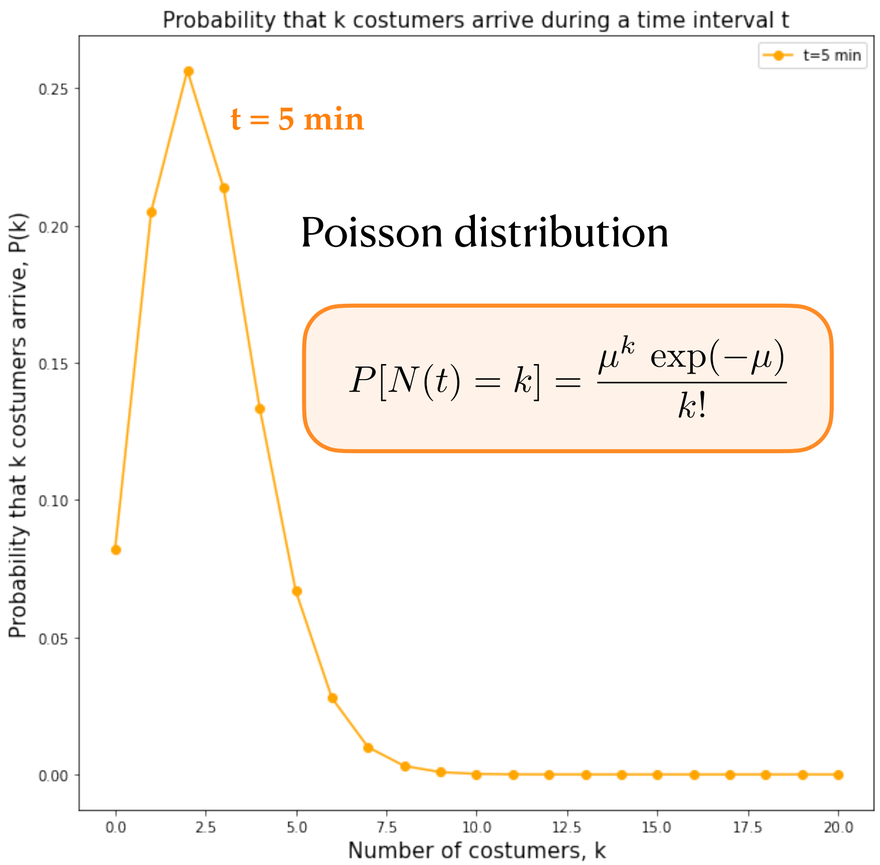
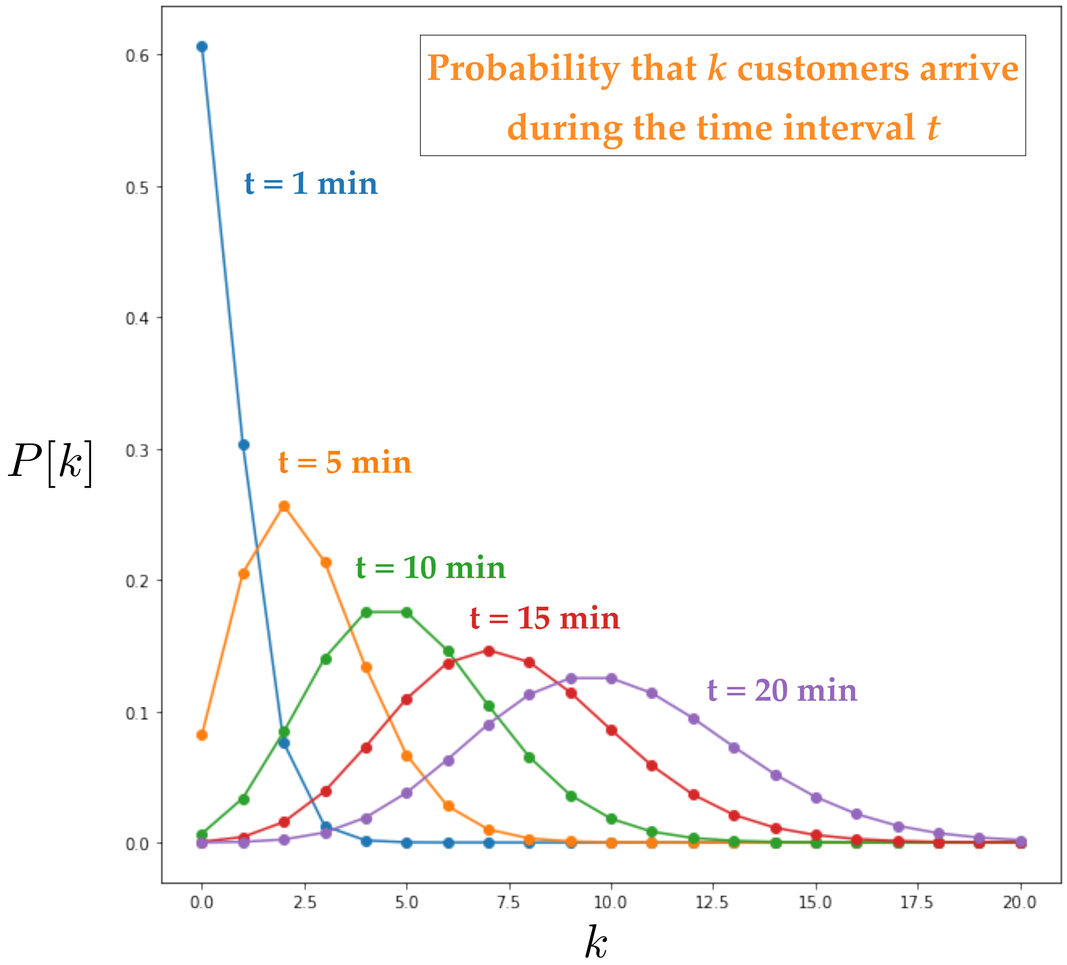
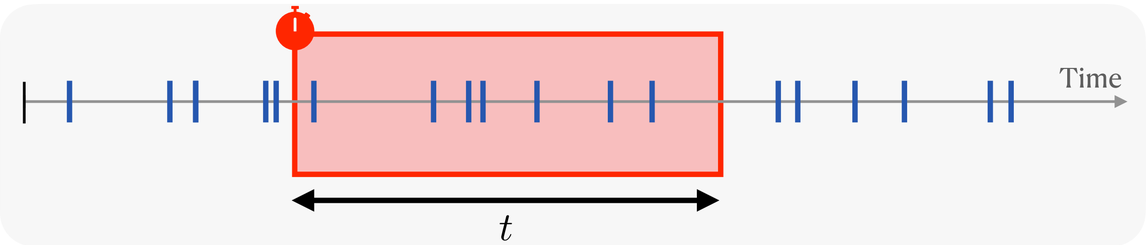


Poisson distribution





Poisson distribution with parameter $\mu = \lambda t$

$$P[N(t) = k] = \frac{\mu^k \exp(-\mu)}{k!}$$

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Expansion for very short time interval dt

Very short time interval dt

Very short time interval dt

The counting process $N(t)$ is a **Poisson process** with rate $\lambda > 0$, if:

- $N(0) = 0$
- $N(t)$ has independent increments;
- during a very short time interval dt :
  - the probability that 1 customer arrives is $\lambda\, dt + o(dt)$
  - the probability that no customer arrives is $(1 - \lambda\, dt) + o(dt)$
  - the probability that 2 or more customers arrive is $0 + o(dt)$

Very short time interval dt

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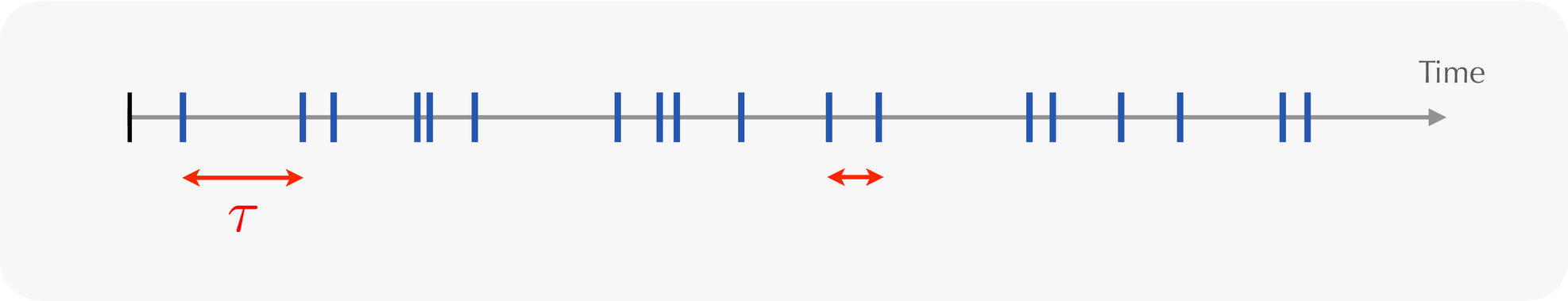
—> Will be very useful for working with continuous-time Markov processes

Distribution of the waiting time

What is the probability that the next customer arrives in the next 1 min?

Distribution of the waiting time



τ = waiting time between the arrivals of two successive customers

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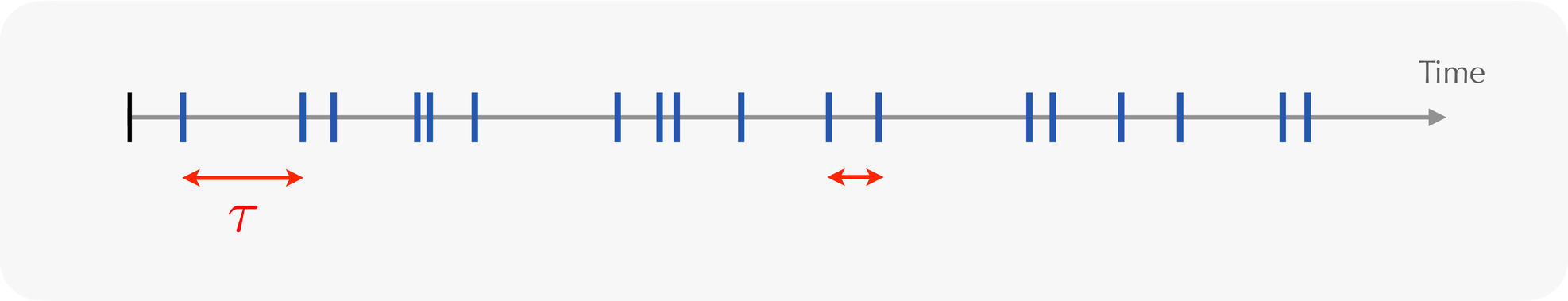

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$$P(\tau)$$

$$P(\tau)\, d\tau$$

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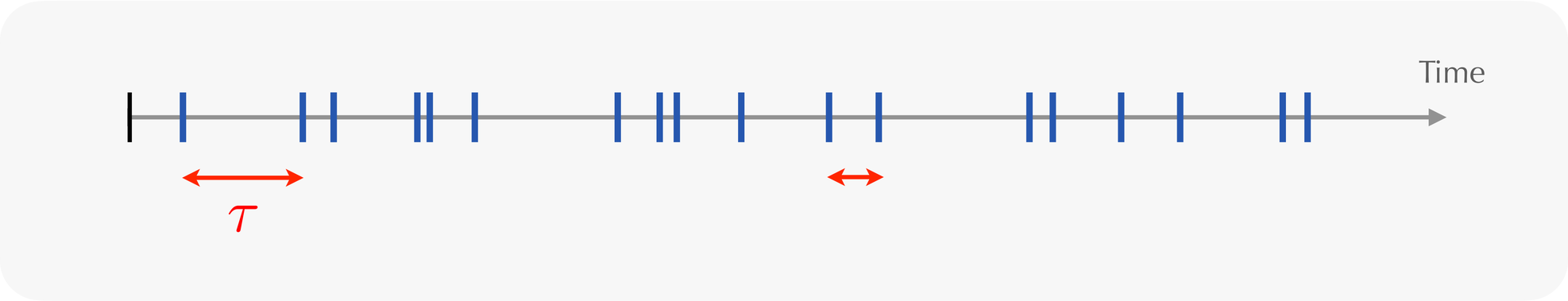


τ = waiting time between the arrivals of two successive customers

$P(\tau)$ = probability distribution of the waiting times

$P(\tau)\,d\tau$ = probability that a customer arrives between during a very short time interval $[\tau,\ \tau + d\tau)$

What is the probability that the next customer arrives in the next 1 min?

$$U(T) = \int_{\tau=0}^{\tau=T} P(\tau)\, d\tau$$

"Cumulative"

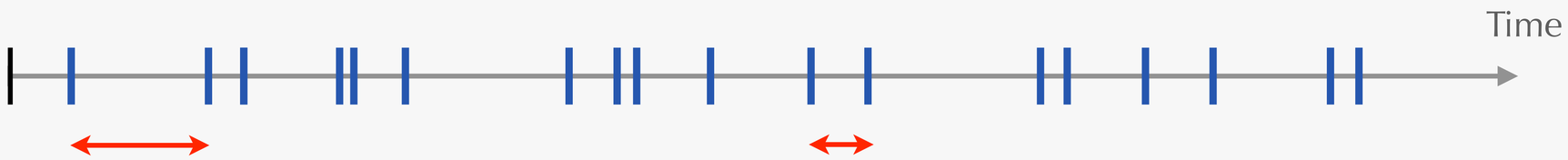
What is the probability that the next customer arrives in the next 1 min?

$$U(T) = \int_{\tau=0}^{\tau=T} P(\tau)\, d\tau$$

= probability that the customer arrives at any time between the time $\tau = 0$ ("now") and the time $\tau = T$

"Cumulative"

Distribution of the waiting time



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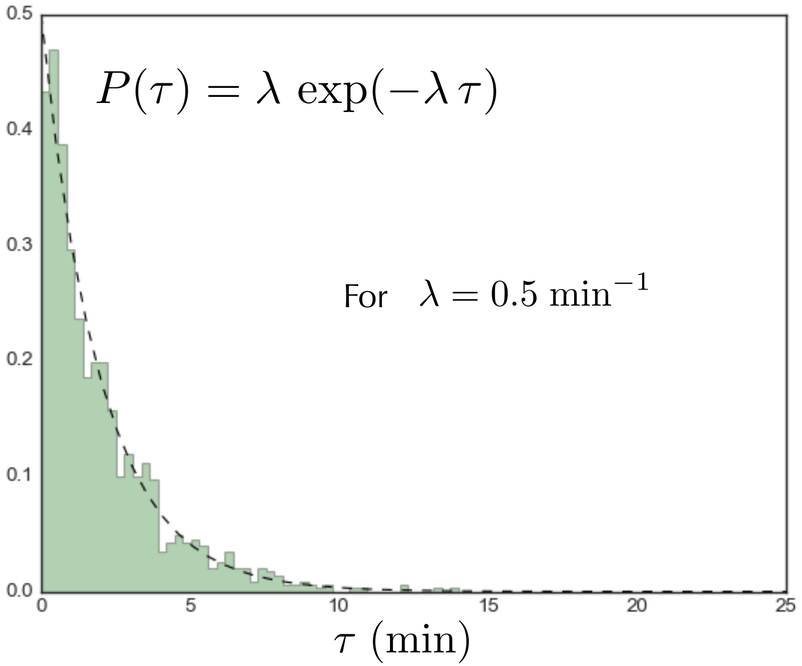
Exponential distribution

$$P(\tau) = \lambda \exp(-\lambda \tau)$$



Poisson Processes

Summary

Poisson Process — Summary

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Occurrence of events:

- Random
- Independent
- With constant rate λ

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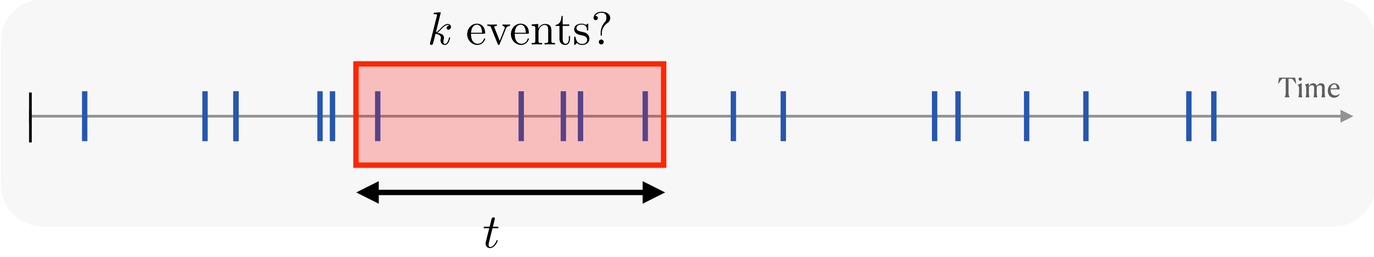

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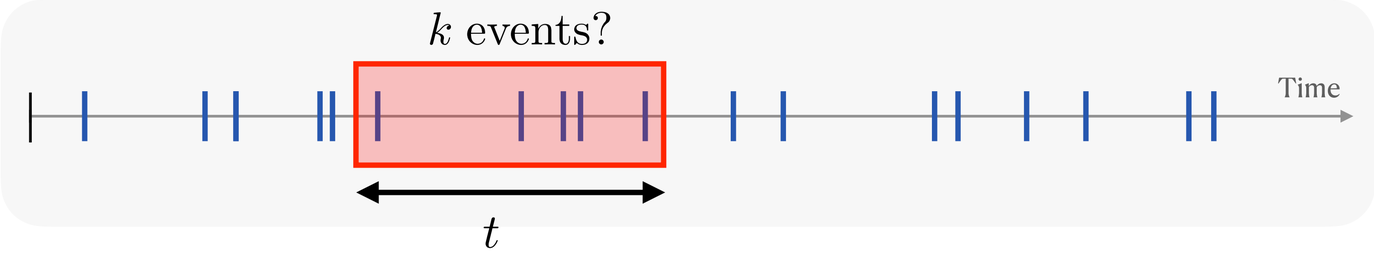

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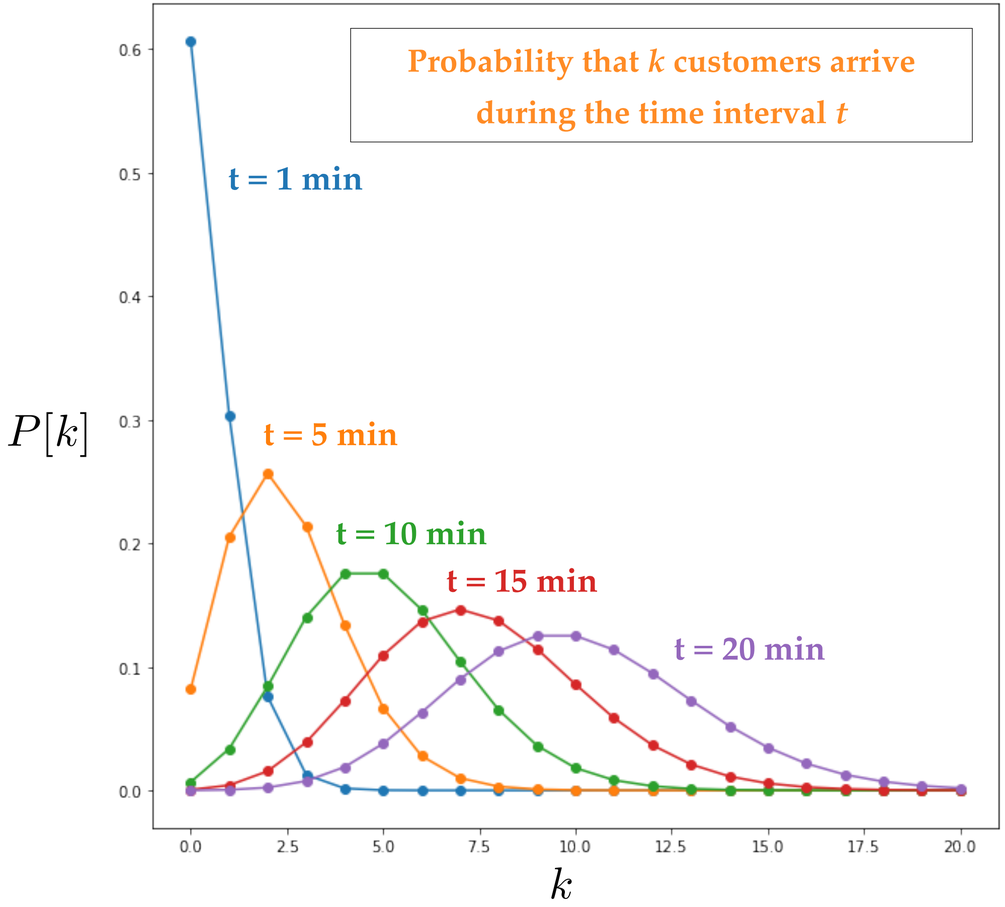

Ex. Answer the question:

What is the probability that exactly 2 customers arrive within 5 min?

Poisson Process — Summary

Occurrence of events:

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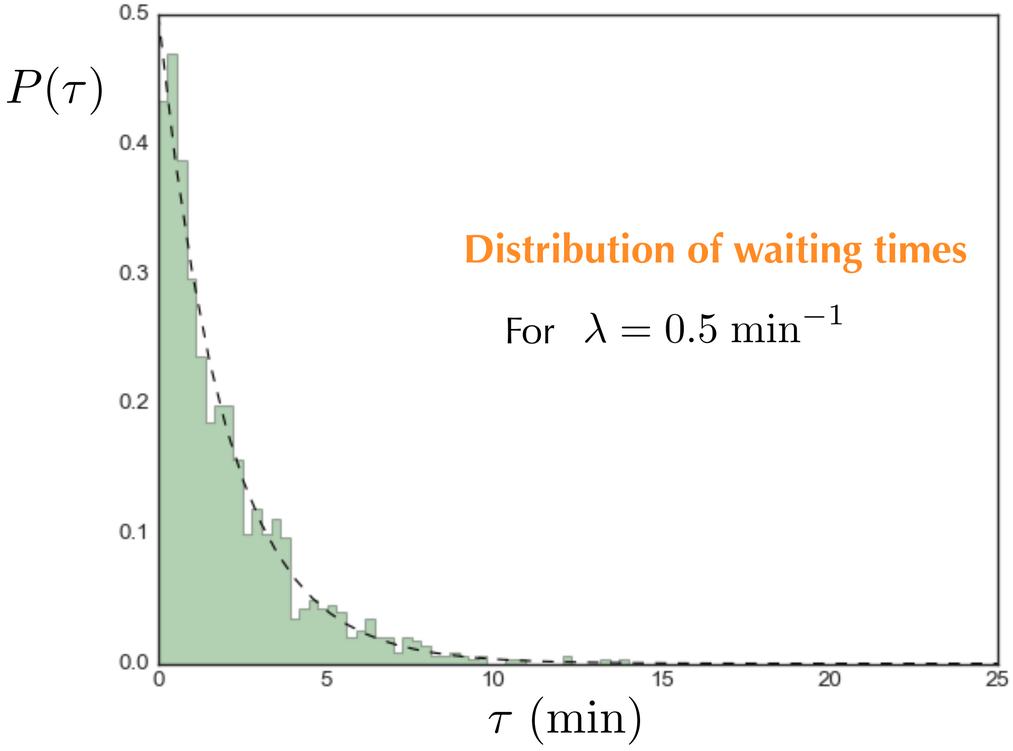

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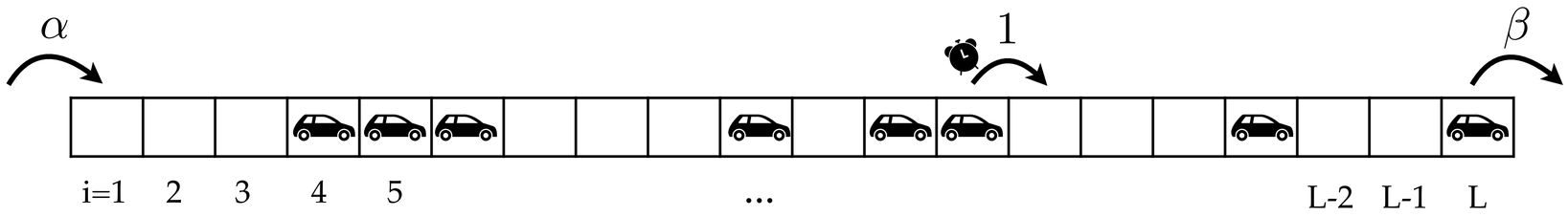
What is the probability that the next customer arrive within 1 min?

Applying what we have just seen to another model…

TASEP

A Simple Traffic Model

Totally Asymmetric Simple Exclusion Process (TASEP)



"**Totally Asymmetric**": cars can only move towards the right

"**Simple**": move only by one site at a time

"**Exclusion**": there can only be one car per site

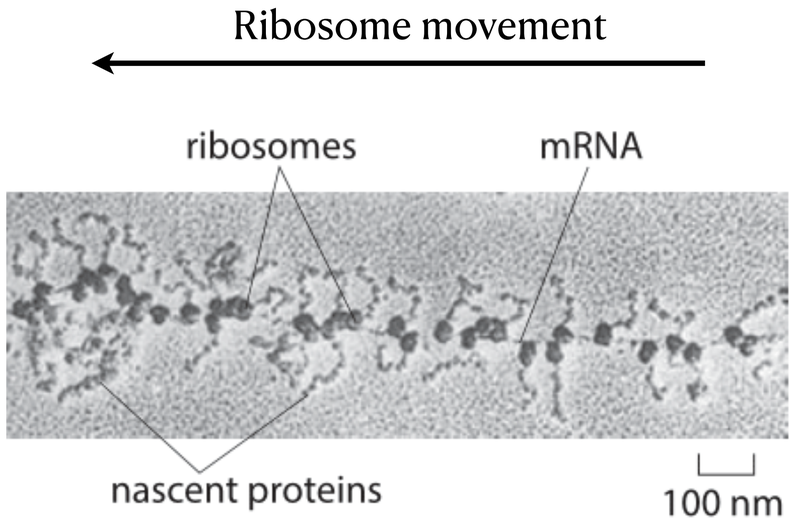
α = **rate** at which cars **enter from the left** of the lattice

β = **rate** at which cars **exit from the right** of the lattice

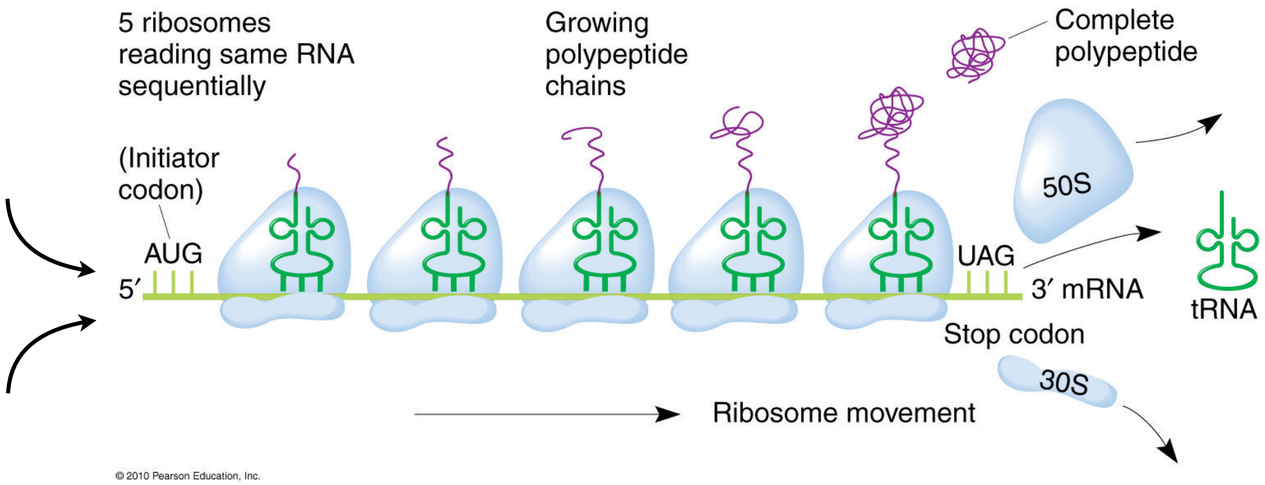
1 = **rate** at which cars **move to the right next site**

This only happens if the next site is free!

Transcription of mRNA



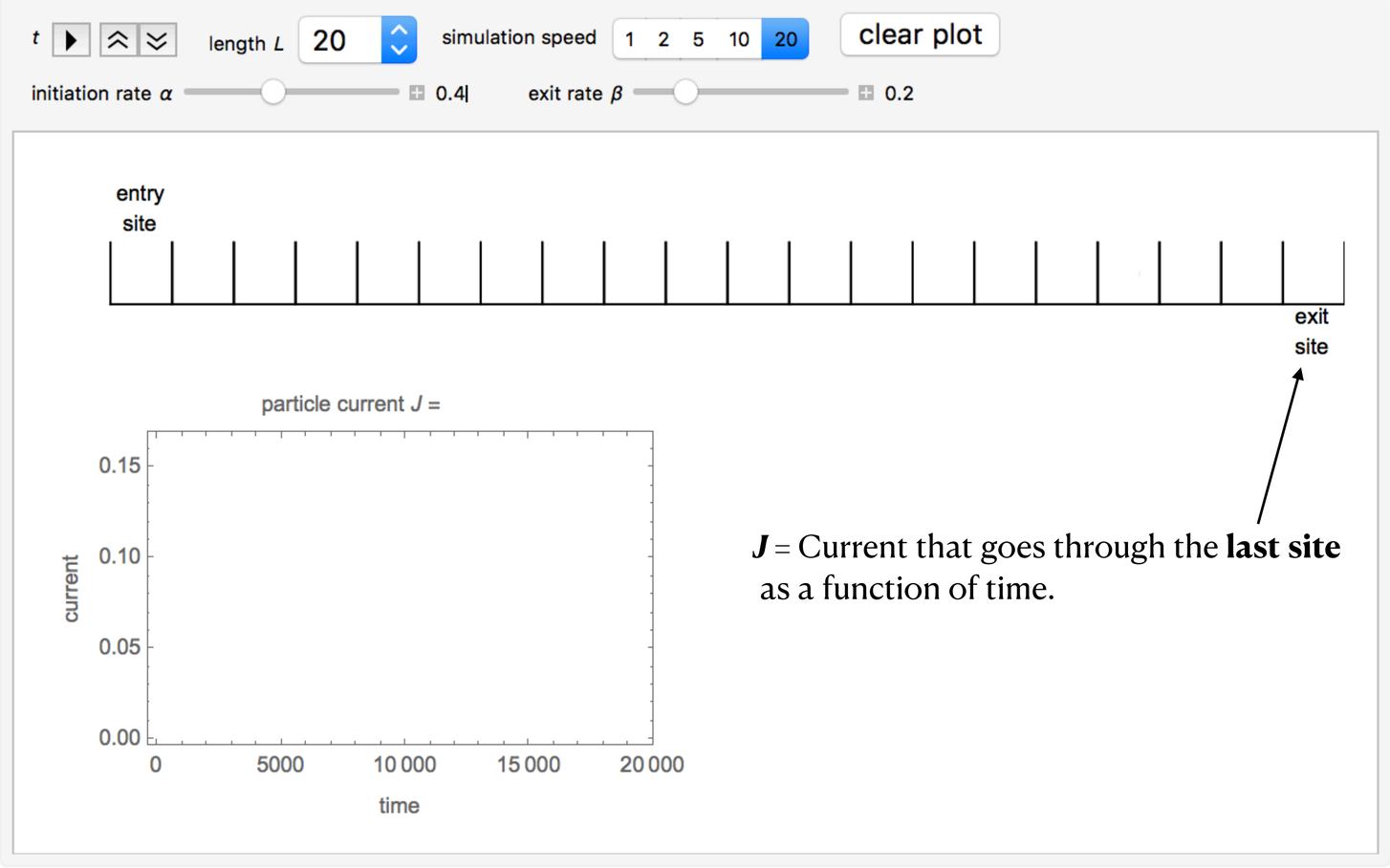
(polypeptide chain gets longer as ribosome moves to the exit)



Schematic drawing of a polyribosome (polysome)

Simulation

Simulation written in Mathematica: you can find how to get a UvA student licence and install it here



J = Current that goes through the **last site** as a function of time.

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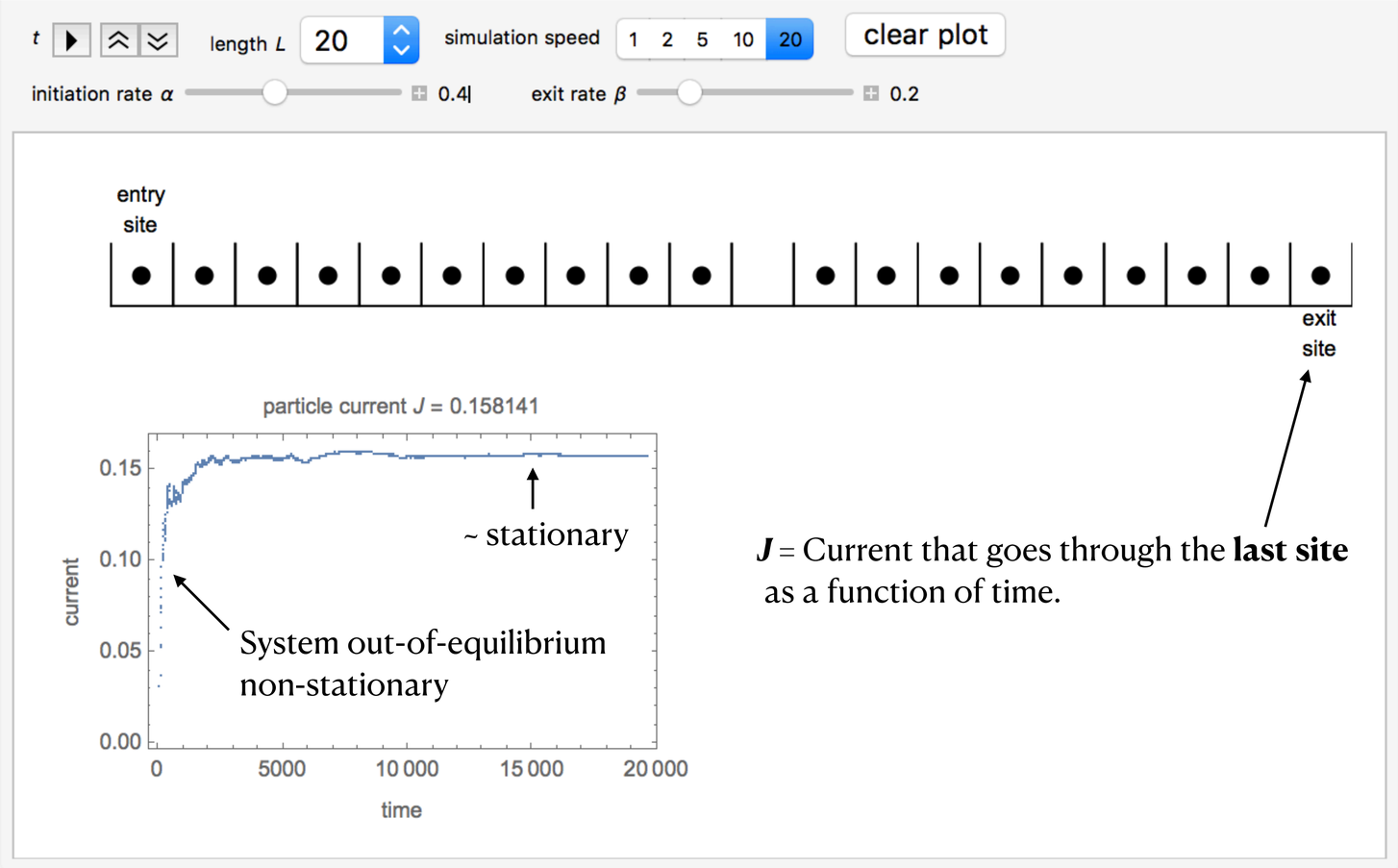


J = Current that goes through the **last site** as a function of time.

To do before lecture on Wednesday:

Play with the simulation:

— initially out-of-equilibrium

— then stationary

Behaviour of the system for

$(\alpha,\ \beta) = (1,\ 0.1)$? $(\alpha,\ \beta) = (0.1,\ 1)$?

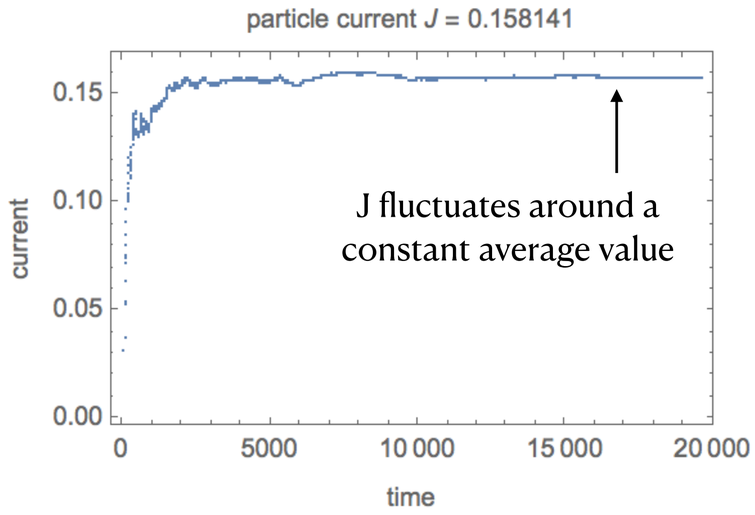
$(\alpha,\ \beta) = (0.2,\ 0.2)$? $(\alpha,\ \beta) = (0.6,\ 0.6)$?

Which case produce a traffic jam pattern similar to what can be seen on this video?
(link here)

For Wednesday

— **Finish the exercise** —> will be corrected during the tutorial session

— To do **before the lecture on Wednesday:** Fill in [this table] using the simulation on Mathematica **~2 values per person!**



beta \ alpha	0,1	0,2	0,3	0,4	0,5	0,6	0,7	0,8	0,9	1,0
1,0										
0,9									0,27	
0,8										
0,7										
0,6	0,09									
0,5				0,24						
0,4										
0,3										
0,2										0,16
0,1										

Advise for the course:

—> work in groups 2-3 people max.
with mix background (theory, computational, CS, …)

Please contact us if you have any difficulty joining a group!